HENAN VARGHESE

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SKILLS

- **Languages**: C, C++, Python
- **Frameworks**: Arduino Framework, MATLAB/Simulink
- **Tools/Platforms**: PTC Creo, Proteus, Keil µVision, Cisco Packet Tracer
- **Soft Skills**: Problem-Solving, Microsoft Office Applications, Circuit design and simulation

INTERNSHIP

- **NextGen Networking: Summer Internship in Advance Computer Networks** June–July 2024

  - **About**: Completed an internship focused on networking, covering OSI and TCP/IP models, subnetting, routing and switching, VLANs, and network security. Gained practical skills in configuring devices and simulating real-world scenarios using Cisco Packet Tracer.

PROJECTS

- **Smart Yoga Mat** Mar 2024 - Apr 2024

  - Designed an interactive yoga mat using FSR sensors, LEDs, buzzer, and LCD to detect and provide feedback on four yoga postures.
  - Developed ESP32-based system for real-time posture validation with asana-specific pressure thresholds and intuitive user interface.
  - Programmed logic for visual (LED) and audio (buzzer) alerts to guide users towards correct posture alignment.
  - Integrated 4 push buttons for asana selection and 20x4 LCD to display posture status, instructions, and feedback clearly
  - Applied embedded systems, sensor interfacing, and health-tech concepts to enhance safe and accurate yoga practice

  Tech: Embedded System, Arduino IDE

- **Energy Monitoring System** Mar2024- Apr 2024

  - Created an energy monitoring system using the 8051 microcontroller to measure voltage, current, power, and energy consumed by electrical loads.
  - The hardware used includes - 8051 microcontroller, ADC0804, voltage sensor, current sensor (ACS712/CT), 16x2 LCD, and regulated power supply.
  - The softwares and tools used were Keil µVision for embedded C programming and Proteus for circuit design and simulation.
  - Applications – Useful for domestic and industrial energy monitoring, smart meters, and efficient power management systems.
  - **Tech**: Embedded System, C programming

CERTIFICATES

- NextGen Networking – Centre of Professional Enhancement, LPU June – July '25
- The Complete Python Developer - Udemy Mar '24
- Python course for beginners at Scalar Sep'23
- Foundation course on Embedded C programming using STM32 Microcontroller at Udemy Oct'23

EDUCATION

- **Lovely Professional University** Punjab, India
  Bachelor of Technology – Electronics and Communications Engineering; ***CGPA: 6.67*** *Since August 2023*
- **HILL BLOOMS SCHOOL** , Kerala
  Intermediate; *April 2022 - March 2023*
- **HILL BLOOMS SCHOOL** Kochi, Kerala
  Matriculation; *April 2020 - March 2021*